

Lecture 33

2 sided inverse = inverse

$$AA^T = I = A^T A$$

$m = n = r$ $\rightarrow$ condition for this.

Left inverse

$\Rightarrow n = r$ (full column rank)

$\Rightarrow$ Nullspace = zero vector

$\Rightarrow$ 0/1 soln for $Ax = b$

$\Rightarrow A^T A = n \times n$ square matrix
 $r = n$ [full rank!!]
invertible!

Now we can say that

$$(A^T A)^{-1} A^T \rightarrow \text{Left inverse.}$$

$$\text{Because } \rightarrow \underbrace{(A^T A)^{-1} A^T}_A A = I$$

A_{left}^{-1}

$\Rightarrow$ Right inverse

- Full row rank

$$\underline{r=m} < \underline{n}$$

- Nullspace is at least 1 dimensional.

- Many soln for $Ax=b$

$AA^T \rightarrow m \times m$ symmetric matrix
rank = m
full rank.

$A^T(AA^T)^{-1} \rightarrow$ Right inverse.

as
$$\underbrace{A(A^T(AA^T)^{-1})}_{A_{\text{right}}^{-1}} = I$$

What do we get when we
do $\rightarrow AA_{\text{left}}^{-1}$ & $A_{\text{right}}^{-1}A$

$$\Rightarrow AA_{\text{left}}^{-1} = A \underbrace{(A^T A)^{-1} A^T}_{\downarrow} = P$$

Projection onto
column space of A

$$\Rightarrow A_{\text{right}}^{-1}A = \underbrace{A^T (A A^T)^{-1}}_{\downarrow} A$$

Projection onto
row space of A

Pseudo inverse

⇒ There are cases when neither left or right inverses exist

→ The case when there is
 $r < n$ & & $r < m$

→ There is nullspace for both
 A & A^T .

⇒ One thing before pseudoinverses →

There is a one-to-one mapping from row space to column space.

So, by limiting A from row space to column space, it is invertible. It is called pseudo-inverse.

Prove

If x, y are in row space of A then $Ax \neq Ay$ ($x \neq y$)

$\Rightarrow$ Suppose $Ax = Ay$

$$\Rightarrow A(x-y) = 0$$

So, $x-y$ must be in the nullspace of A but we know that it is in the row space.

Hence we have a contradiction!

$$\boxed{x = y}$$

⇒ How to find pseudo inverse A^+

① Start from SVD

$$A = U \Sigma V^T$$

$$\begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_r & & 0 \end{bmatrix}_{m \times n}$$

$$\Sigma^+ = \begin{bmatrix} 1/\sigma_1 & & & \\ & 1/\sigma_2 & & \\ & & \ddots & \\ & & & 1/\sigma_r & & 0 \end{bmatrix}_{n \times m}$$

$$\Sigma \Sigma^+ = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 & & 0 \end{bmatrix}_{n \times m}$$

$$A^+ = V \Sigma^+ U^T$$